

MACHINE LEARNING MODEL — COMPLETE TECHNICAL DOCUMENTATION

MastitisGuard AI

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Problem Statement Number: 26109

Journey stage: Built solution

1. Introduction

Bovine mastitis is a leading cause of economic loss in Indian dairy farming, largely because it remains invisible until milk yield has already dropped. Team H14 SPIRIT built MastitisGuard AI to close that gap: a three-unit Edge AI system — an animal wearable, a milk measurement unit, and a central Shed processing unit — that identifies every cow individually, learns that cow's own normal behaviour and milk pattern, and forecasts mastitis risk ahead of clinical symptoms.

The predictive engine has two layers. The first is a deterministic per-cow baseline layer: a rolling median of each cow's own recent healthy readings, always active, requiring no training data, and running on the Shed ESP from day one of deployment. The second is a CatBoost-based risk classifier trained on top of that baseline data, which sharpens the raw deviation signal into a single, farmer-readable risk tier. This document covers both layers in full: Section 3 covers per-cow identification, Section 4 covers how the baseline layer is computed, and the remaining sections cover how the CatBoost layer is trained, evaluated, deployed offline, and designed to keep improving over time.

2. Purpose of This Document

This document is the technical reference for the MastitisGuard AI predictive model specifically. It is intended to let anyone — a teammate, a judge, or a future contributor — understand not just what the model does, but why each design choice was made, so that every number and result in this project can be defended rather than taken on faith. Every design number below is expanded on, with its specific justification, in the Validation & Justification document.

3. Unique Cow Identification (Barcode / QR System)

Every reading in the system — from the wearable, the milk unit, or a CMT test — must be attributable to one specific animal, or a per-cow baseline is meaningless. This is achieved through a unique Cow ID assigned to every animal at registration, printed as a barcode/QR tag on an ear tag.

How Identification Works in Practice

- At registration, each cow is assigned a unique Cow ID (e.g., COW-023) and fitted with an ear tag printed with a corresponding barcode/QR code.
- The wearable collar continuously tags every sensor packet it sends with this Cow ID, so behavioural data is never anonymous.
- At the milking point, an ESP32-CAM with an OV2640 camera scans the ear tag's barcode/QR code before milk parameters are recorded, automatically linking that session's milk temperature, pH, yield, and CMT result to the correct animal.

- The Shed ESP (central Edge AI unit) uses this ID to route incoming data to the correct per-cow history buffer — wearable data and milk data for the same cow are only ever combined with each other, never mixed across animals.

4. Cow-Specific Baseline — A Core Strength of This Solution

Cow A is naturally calmer and less active than the herd average — this is just her temperament, not illness. Cow B is naturally very active. A single fixed activity threshold applied to the whole herd would either flag Cow A as "sick" every single day (a constant false alarm the farmer learns to ignore), or set the threshold so low that a genuine slowdown in Cow B's activity — an early sign of illness — never crosses it at all (a missed detection). Comparing each cow only to her own recent history avoids both failure modes at once — and this baseline layer runs today, on-device, independent of the CatBoost layer described later in this document.

How the Baseline Is Computed

For every sensor channel, the system maintains a rolling historical buffer across that specific cow's last 7 to 14 days of verified healthy readings, computing two derived layers:

Layer	Feature Nomenclature	Definition
Raw Reading	raw_<feature>	Instantaneous value sampled from collar wearable or milking node.
Personal Rolling Baseline	roll7_<feature>	Cow-specific moving median (7–14 day window) representing her individual healthy normal.
Deviation Residual	dev_<feature> = raw_<feature> – roll7_<feature>	Relative drift from the cow's own baseline, eliminating genetic/temperament offsets.

The CatBoost layer (Section 6) is trained primarily on directional deviation features (dev_*) rather than absolute values (raw_*), because a relative departure from an animal's personal healthy baseline is a stronger, more specific physiological indicator of subclinical disease than any static herd-wide threshold.

Physiological Grounding Used

The healthy physiological baselines and subclinical drift windows used to construct, normalize, and validate this dynamic rolling baseline (7–14 days) are grounded in published clinical trials and veterinary literature (see Team H14_SPIRIT's Research and References list).

Signal	Healthy Range	Mastitis Effect	Sensor Status
Skin/contact temperature	38.0–38.5 °C	Rises 0.3–1.0 °C	Built (MAX30205 specified; DHT11 bench stand-in — see Bill of Materials)
Rumination	400–500 min/day	Drops 40–100 min/day	Built (MPU6050 proxy)
Activity	Individual per-cow baseline	Drops 15–35% (RMS motion)	Built (MPU6050)
Milk pH	6.5–6.8	Shifts alkaline to 6.9–7.4	Budgeted, not yet physically integrated (see Bill of Materials, Section 3)
Milk electrical conductivity	4.0–5.5 mS/cm	Rises to 5.8–6.8 mS/cm	Synthetic-data feature only — no physical EC probe is budgeted or

Signal	Healthy Range	Mastitis Effect	Sensor Status
			installed yet; retained in the schema for a Phase 2 EC probe addition
Milk temperature	34.5–35.9 °C	Rises 0.2–0.8 °C	Built (DS18B20)
Milk yield	Individual per-cow baseline	Drops 0.5–2.5 kg/session	Built (load cell + HX711)
CMT score	Negative (0)	Rises to Trace/+/++/+++ once clinical	Manual paddle built; used only as a validation label, never a model input feature (Section 5)

5. Dataset and Prediction Target

The model predicts `label_pre_clinical_mastitis`: will this cow show a clinical/CMT-positive mastitis reading within the next 1–5 days? This is explicitly an early-warning target, not a "does this cow currently have mastitis" classifier, and it is the demonstrated benchmark referenced everywhere else in this project's documents — the 7–14 day figure quoted in the Technical Dossier is the longer-term literature-supported target this project is working toward, not a claim about this dataset.

400 simulated cows across 20 days each (8,000 readings) were used for prototype validation, grounded in the physiological ranges above. Around 30% of cows develop mastitis during the simulated window; each such cow has a 5-day sub-clinical drift window immediately before the clinical onset day, during which sensors gradually shift — this window is exactly what is labelled positive for training.

Critical design decision: CMT results and the future camera-based CMT reading are deliberately excluded from the model's input features. CMT only turns positive once mastitis is already clinical — including it as an input would let the model simply learn to echo a test that is already confirming the disease, defeating the purpose of early warning. Excluding it forces the model to learn genuine early behavioural and milk-quality drift instead.

6. Model Architecture and Training

Why CatBoost

- Handles categorical features (breed, lactation stage, season) natively, without manual encoding.
- Has a native exporter to standalone C++, which is what makes fully offline Edge AI deployment on the ESP32 possible without a separate model-conversion step.

Training Configuration

- 400 trees maximum, depth 6, learning rate 0.05.
- Class weights of [1, 12]: only around 8% of readings are genuinely positive, matching real mastitis incidence. Without this weighting, the model could achieve high apparent accuracy simply by always predicting "healthy"; the 12x weighting forces it to actually learn the rare early-warning pattern.
- Train/test split performed by individual Cow ID, not by row: this prevents the model from being tested on the same animal it was partly trained on, which would let it "memorize" that cow's baseline instead of learning a generalizable pattern.

7. Results — Synthetic Data Only

All figures in this section are model performance on held-out synthetic cows. **They are not a claim of validated real-world field accuracy** — that validation is the defined Phase 1 milestone described in Section 11 and in the Decision-Making and PoC document, Section 4.

Metric	Value
ROC-AUC (held-out synthetic cows)	0.964
F1 score (tuned threshold)	0.75
Precision (mastitis class)	0.81
Recall (mastitis class)	0.70
Decision threshold	0.729
Fusion weight — tabular sensors	1.00 (100%)
Fusion weight — vision/CMT confirmation	0.00 (0%)

The fusion-weight result is the most important finding to highlight: when the tabular sensor model is combined with the vision/CMT confirmation signal and the blend is tuned on **validation data** rather than fixed in advance, the optimal blend **places 100% of the weight on the tabular sensors and 0% on vision/CMT** for the early-warning task. Behavioural and milk-quality sensors forecast early; CMT — whether read manually or by future camera automation — **confirms after the fact**. The data itself reveals this; it was not **assumed in advance**.

8. Explainability

A **SHAP (SHapley Additive exPlanations)** summary was generated to show which features drive each individual prediction. This directly supports the "Smart Risk Scoring" innovation claimed in the project's Solution Overview — converting complex multi-sensor data into a risk tier the farmer can act on, while still being able to show a veterinarian exactly which readings triggered the alert.

9. Risk Tiers and Recommended Farmer Action

Risk Tier	Trigger Condition	Recommended Action
SAFE	Fused risk score below half the decision threshold (< 0.365)	Continue normal daily monitoring. No action required.
MEDIUM RISK	Fused risk score between 0.365 and 0.729	Increase monitoring frequency; repeat the milk check at the next session; watch for continued deviation over the next 1–2 days.
CRITICAL	Fused risk score at or above 0.729	Immediate alert to the farmer; contact a veterinarian for physical examination before symptoms become visible.

10. Offline Edge AI Deployment

The trained **CatBoost model is exported via its native C++ exporter into a single self-contained file** with no external dependencies, embedding all decision trees directly. This file is compiled directly into the **Shed ESP's firmware**, so the trained model — evaluated to this point on synthetic data — runs with zero internet

connection and no cloud API call, directly satisfying the requirement for reliable operation on farms with poor or no connectivity.

The feature ordering used by the firmware (21 numeric features, 3 categorical features) was verified directly against the exported model's **embedded metadata** before deployment, rather than assumed — an ordering mismatch here would produce confident-looking but meaningless predictions with no error thrown, so this was checked explicitly rather than trusted by convention.

11. Self-Learning Architecture (Continuous Improvement Design)

The model deployed on-device is a fixed, offline snapshot at any given time — an ESP32 cannot practically retrain a **gradient-boosted model on-device**. "Self-learning" in this system is therefore designed as a **periodic feedback loop** between the field deployment and a central training pipeline, not as continuous on-device learning. This distinction is important to state accurately.

The Feedback Loop

- Every CRITICAL or MEDIUM RISK alert is logged with its full feature snapshot (Cow ID, timestamp, all sensor readings and derived deviations).
- When a veterinarian examines the flagged cow, the outcome (confirmed mastitis / false alarm / other condition) is **recorded in the mobile app against that same logged snapshot, closing the loop between prediction and ground truth**.
- These vet-confirmed outcomes accumulate as real, farm-specific training examples — the exact real data the current model lacks, as distinct from the synthetic training set.
- On a periodic schedule (e.g., monthly, or once enough new confirmed cases accumulate), the CatBoost model is retrained centrally on the combined synthetic-plus-real dataset.
- The updated model is re-exported to C++ and pushed to Shed ESP units as a firmware update (over Wi-Fi when available, or manually during a routine farm visit), replacing the on-device model.

Why This Design, Not On-Device Online Learning

On-device online learning (updating model weights directly on the ESP32 as new data arrives) was considered and rejected for this prototype: it would require **retaining raw training data on a memory-constrained microcontroller**, risks the model drifting based on a single animal's unusual readings, and cannot easily incorporate the vet's confirmed diagnosis, which is the actual ground truth needed to improve the model responsibly.

What This Enables Over Time

- Phase 1 (current): model trained entirely on physiologically-grounded synthetic data; this is the stage documented in Section 7.
- Phase 2: first real farm deployment begins logging vet-confirmed outcomes; model is retrained once sufficient real examples accumulate, using the identical feature schema so no code changes are required.
- Phase 3 and beyond: the model becomes progressively more accurate for the specific breeds, climates, and management practices of the farms actually using it, rather than remaining generic.

12. Conclusion

MastitisGuard AI demonstrates that early, **non-invasive mastitis forecasting is achievable** with low-cost, offline-capable hardware, provided the underlying model is built around two disciplined choices: comparing each animal only to its **own history**, and training the model to **forecast** rather than merely confirm. Both choices are reflected directly in the results — an AUC of 0.964 on held-out synthetic cows, and a fusion analysis that independently confirms behavioural and milk sensors carry the early signal, while CMT and vision serve as after-the-fact confirmation rather than early warning. What remains ahead of the current prototype is real-world validation — deploying to an actual herd, collecting vet-confirmed outcomes, and allowing the self-learning feedback loop described in Section 11 to begin operating on genuine farm data. That transition, from a physiologically-grounded synthetic model to a continuously improving farm-specific one, is the project's clearly defined next step, matching the Phase 1 milestone in the Market Research document, Section 4.